12/12

MATHEMATICS 2210
Quiz 8, Fall semester 2012

Name: Key

1. (6 pts) Find the area of the region inside the circle $r = 4\cos(\theta)$ and outside the circle $r = 2$. (Recall that $\int \cos^2(u)du = \frac{1}{2}(\cos(u)\sin(u) + u)$.)

$r = 4\cos\theta \quad r^2 = 4r\cos\theta$

$x^2+y^2 = 4x \quad x^2-4x+y^2=0$

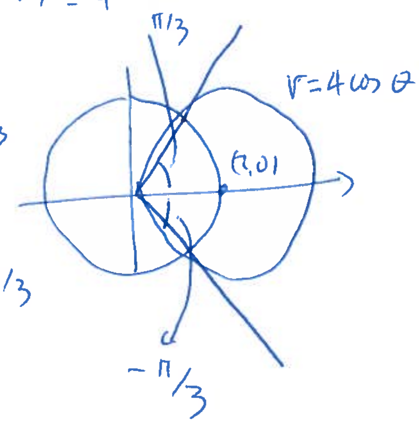
$(x-2)^2+y^2=4$

$$\int_{-\pi/3}^{\pi/3}\int_{2}^{4\cos\theta} r\,dr\,d\theta = \int_{-\pi/3}^{\pi/3} \frac{16\cos^2\theta}{2} - \frac{4}{2}\,d\theta$$

$$= \left[\frac{8}{2}(\cos u\sin u + u) - 2u\right]_{-\pi/3}^{\pi/3} = \left[4\cos u\sin u + 2u\right]_{-\pi/3}^{\pi/3}$$

$$= 4\frac{1}{2}\frac{\sqrt{3}}{2} + 2\pi/3 - \left(-4\frac{1}{2}\frac{\sqrt{3}}{2} - 2\pi/3\right)$$

$$= 2\sqrt{3} + \frac{4\pi}{3}$$



2. (6 pts) Find the volume of the solid above the xy plane, below $z = 2 + x + y$ and between $y = x^2 - 1$ and $y = -x^2 + 1$.

$$\int_{-1}^{1}\int_{x^2-1}^{1-x^2}\int_{0}^{2+x+y} 1\,dz\,dy\,dx$$

$$= \int_{-1}^{1}\int_{x^2-1}^{1-x^2} 2+x+y\,dy\,dx = \int_{-1}^{1}\left[(2+x)y+\frac{y^2}{2}\right]_{x^2-1}^{1-x^2}dx$$

$$= \int_{-1}^{1} 2\left((2+x)(1-x^2)\right)dx = \int_{-1}^{1} -2x^3-4x^2+2x+4\,dx = \left[-\frac{4x^3}{3}+4x\right]_{-1}^{1}$$

$$= -\frac{4}{3}+4+\frac{4(-1)^3}{3}-4(-1) = 8-\frac{8}{3} = \frac{16}{3}$$